



Batterjee Medical College
For Science and Technology

Hormones and their receptors

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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

﴿وَاتَّقُوا اللَّهَ ۖ وَيَعْلَمُ اللَّهُ ۖ وَاللَّهُ

بِكُلِّ شَيْءٍ عَلِيمٌ﴾

Learning outcomes:

By the end of the lecture, the student will be able to:

- **Explain the hormones characteristics and their nature**
- **Describe the Hormones receptor's structure, their distribution, and ligands**

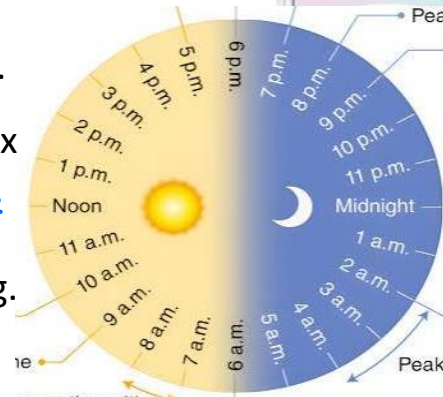
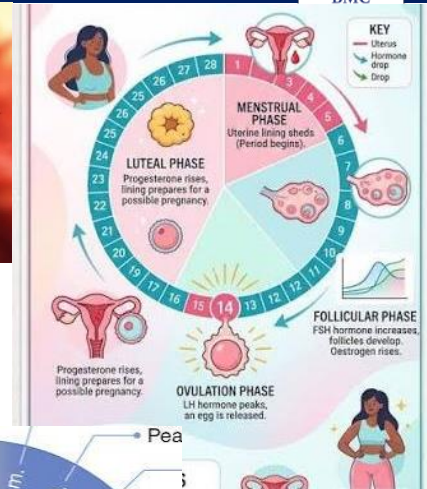


Hormones characteristics and nature.

Hormones characteristics and nature

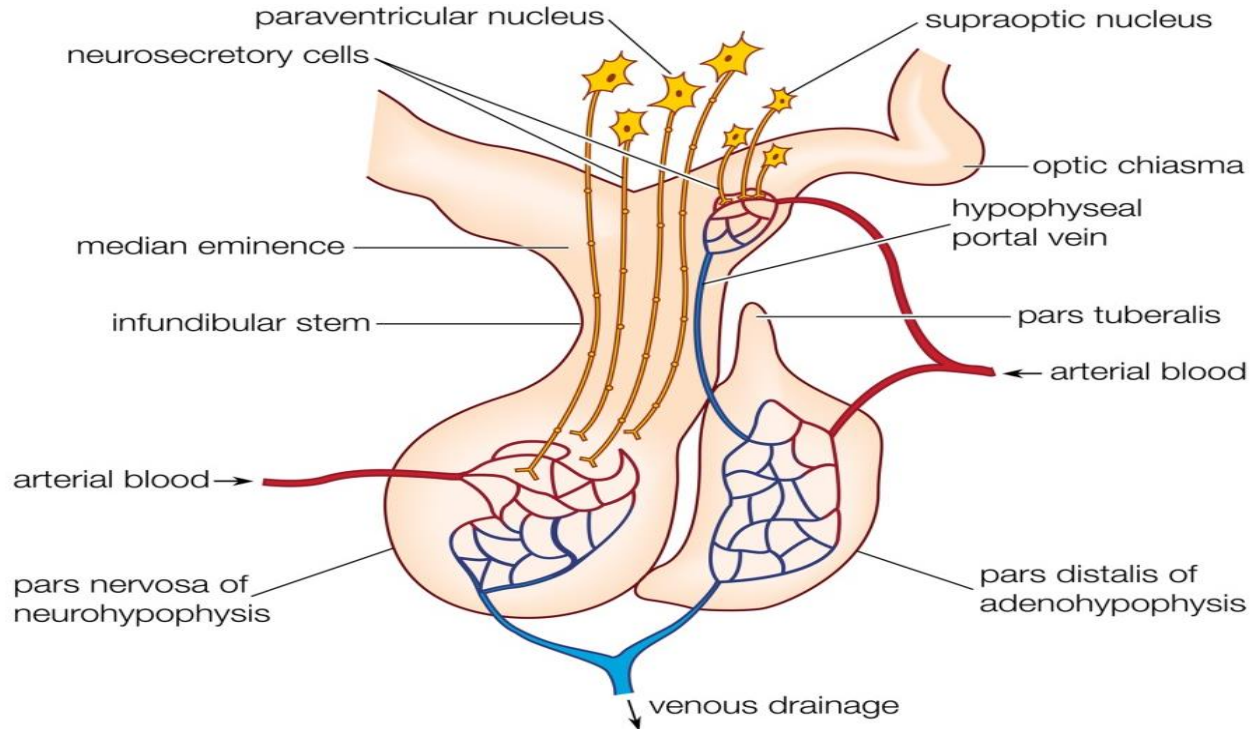
➤ Physiological Properties of Hormones

1. Produced in small amounts (μg to ng).
2. The rate of secretion is determined by the body needs.
3. Super-added rhythms:
 - Circadian rhythm (i.e., daily rhythm), e.g. ACTH.
 - Monthly rhythm, e.g. female gonadotropic hormone.
4. They may affect specific target organs (e.g. sex hormones) or various tissues (e.g. insulin). *Synergize*
5. Some hormones antagonize the effect of others (e.g. insulin and glucagon).



Hormones characteristics and nature

The Inter-relation between the endocrine and nervous systems



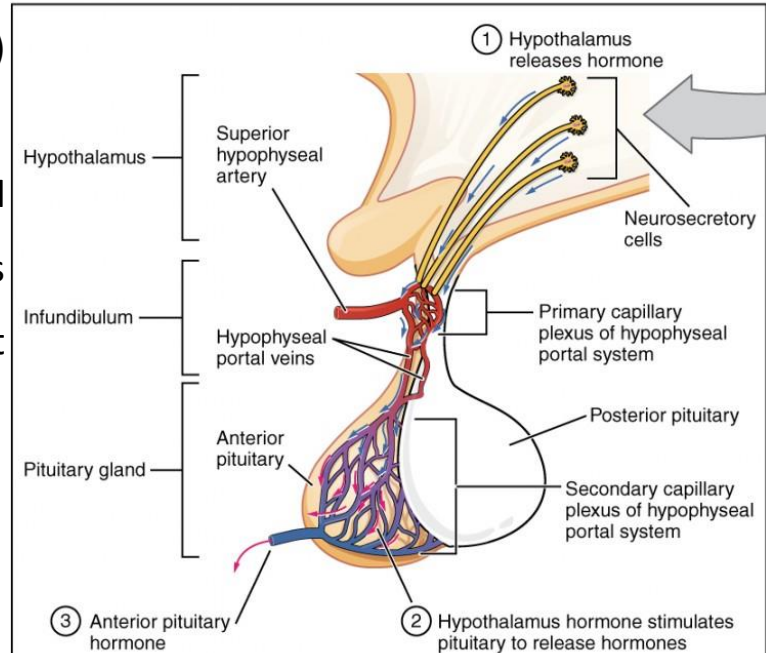
Hormones characteristics and nature

➤ The nervous system can affect the endocrine system:

1. Hypothalamo-hypophyseal portal circulation:

Between the hypothalamus (nervous system) and the anterior pituitary gland.

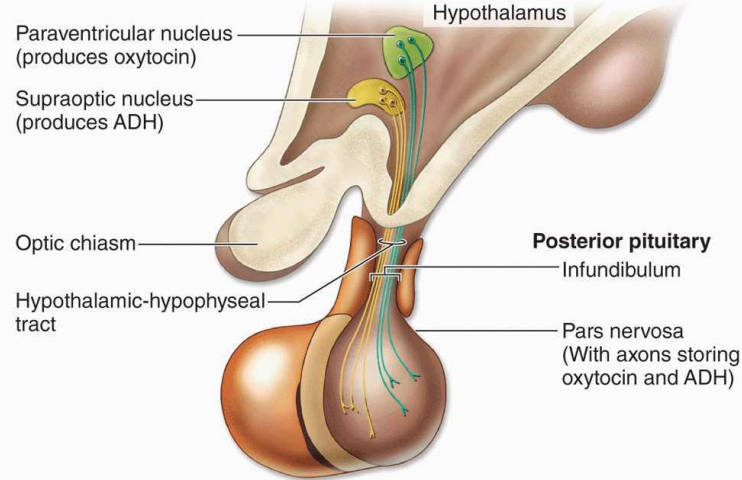
- Releasing or inhibiting hormones are released from the hypothalamus → carried by this circulation → alter the secretion of the different anterior pituitary hormones, e.g., GHRH.



Hormones characteristics and nature

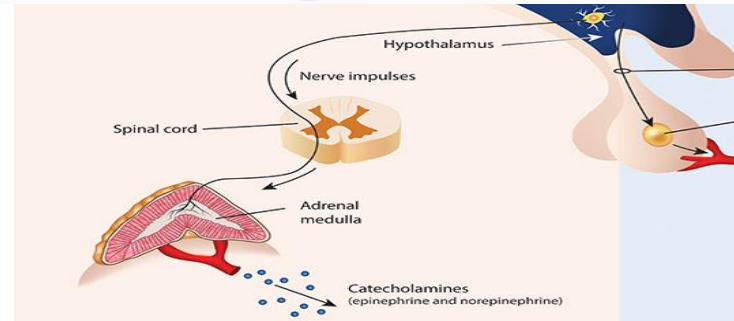
2. **Hypothalamo-hypophyseal tract:** between the hypothalamus and the posterior pituitary.

- The ADH and oxytocin hormones are **formed** in the **hypothalamus**.
- They are **stored** and **secreted** by their nerve endings present in the **posterior pituitary**.



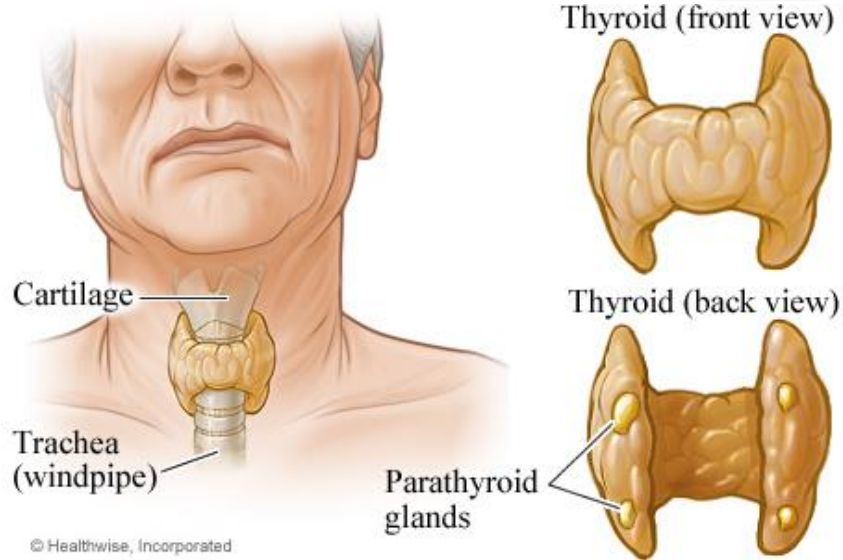
2. **Direct connection between hypothalamus and suprarenal medulla:** Affects the secretion of epinephrine & norepinephrine (NE).

oxyt, ADH



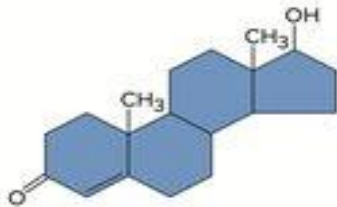
Hormones characteristics and nature

- On the other hand, the endocrine system can also affect the nervous system:
- **Parathyroid** glands control Ca^{2+} level in blood that affects the nerves excitability.
- **Thyroid** hormones affect the excitability of the nervous system.

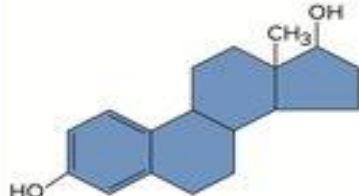


Hormones characteristics and nature

➤ Chemical Nature of Hormones:

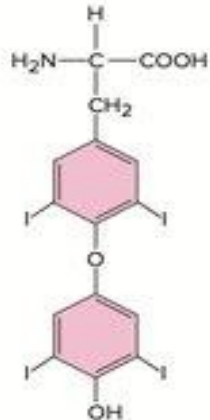


Testosterone

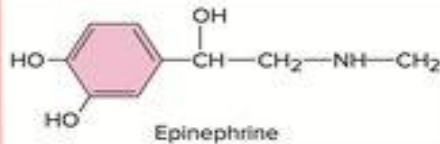


Estradiol

(a) Steroids

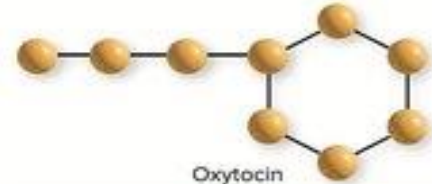


Thyroxine

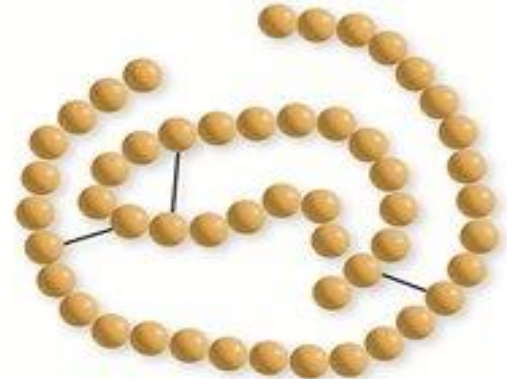


Epinephrine

(b) Monoamines



Oxytocin



Insulin

(c) Peptides

Hormones characteristics and nature

➤ Chemical Nature of Hormones:

- a) **Steroid hormones:** secreted by the **adrenal cortex** (cortisol, aldosterone and androgens), the **placent and ovaries** (estrogen & progesterone) and the **testes** (testosterone).
- b) **Derivatives of the amino acid:** **tyrosine**, secreted by the thyroid (thyroxine and tri-iodothyronine) and the **adrenal medullae** (dopamine, epinephrine and norepinephrine).
- c) **Protein and polypeptide hormones:** including hormones secreted by the **pituitary** gland, the **pancreas** (insulin, glucagon & somatostatin), the **parathyroid** gland (parathyroid hormone), and many others.



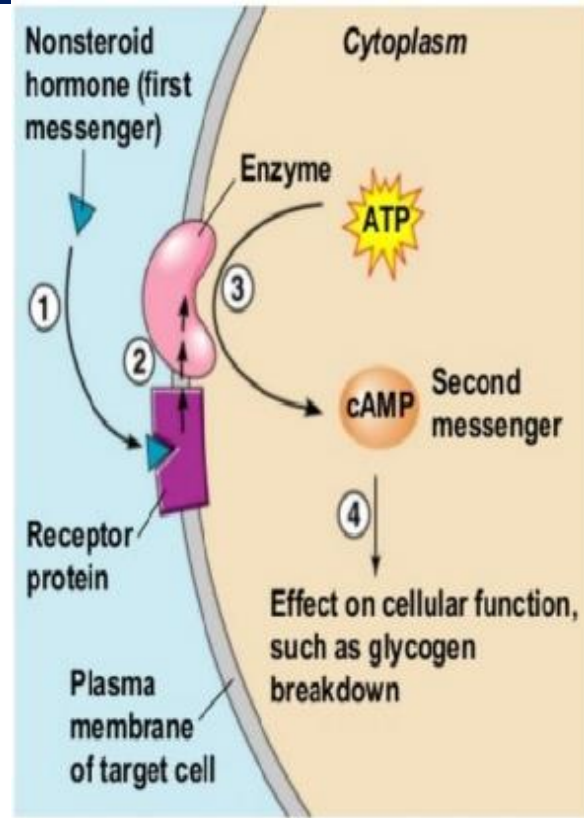
Hormones receptors structure, distribution and ligands

Hormones receptors structure, distribution & ligands

➤ Mechanism of Action of Hormones

- The **first step** of a hormone's action is to bind to specific **RECEPTORS** at the **target** cell.
- Cells that lack receptors for the hormones do not respond (silent).
- The locations for the different types of hormone receptors are generally the following:

1. **In or on the surface of the cell membrane:** Specific mostly for the **protein**, **peptide**, and **catecholamine** hormones.

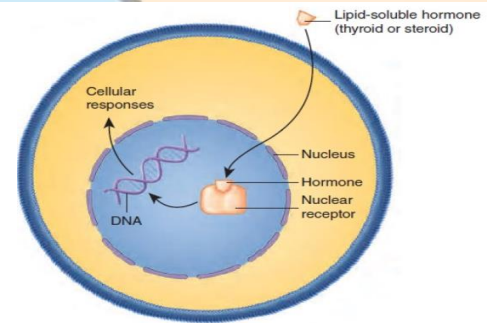
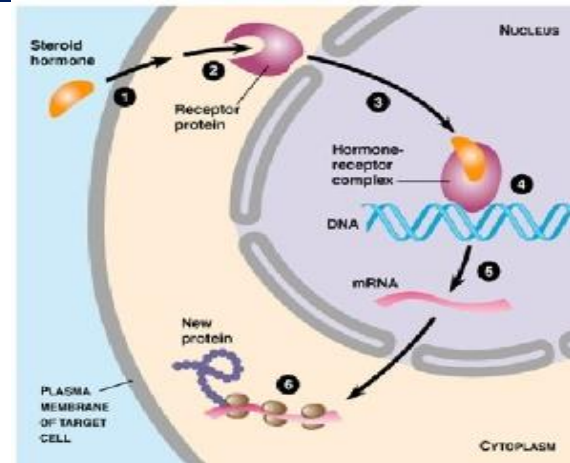


Hormones receptors structure, distribution & ligands

2. **In the cell cytoplasm:** The primary receptors for the different **steroid** hormones.
3. **In the cell nucleus:** Receptors for the **thyroid** hormones are found in the nucleus in direct association with one or more of the chromosomes.

*N.B.: Hormones are called "**primary messengers**", while the chemical substance that results inside the cell from the combination of hormone and receptor is known as the "**second messengers**".*

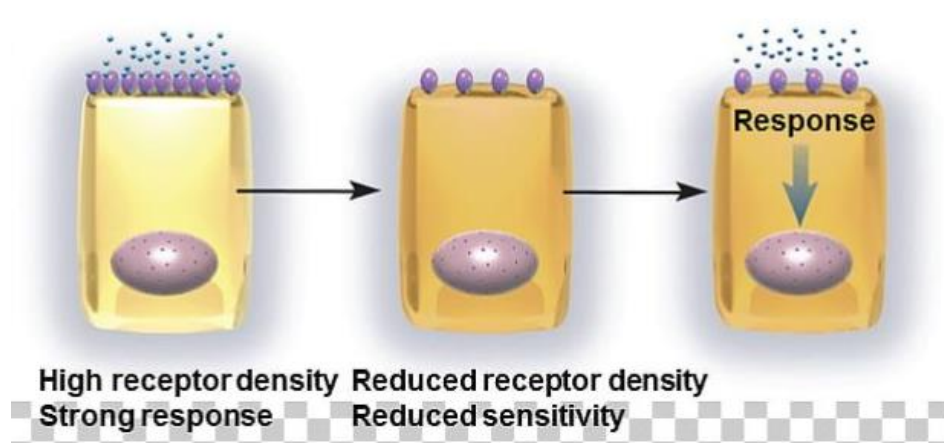
- The 2nd messengers can initiate biochemical reactions that persist after the disappearance of the hormone.



Hormones receptors structure, distribution & ligands

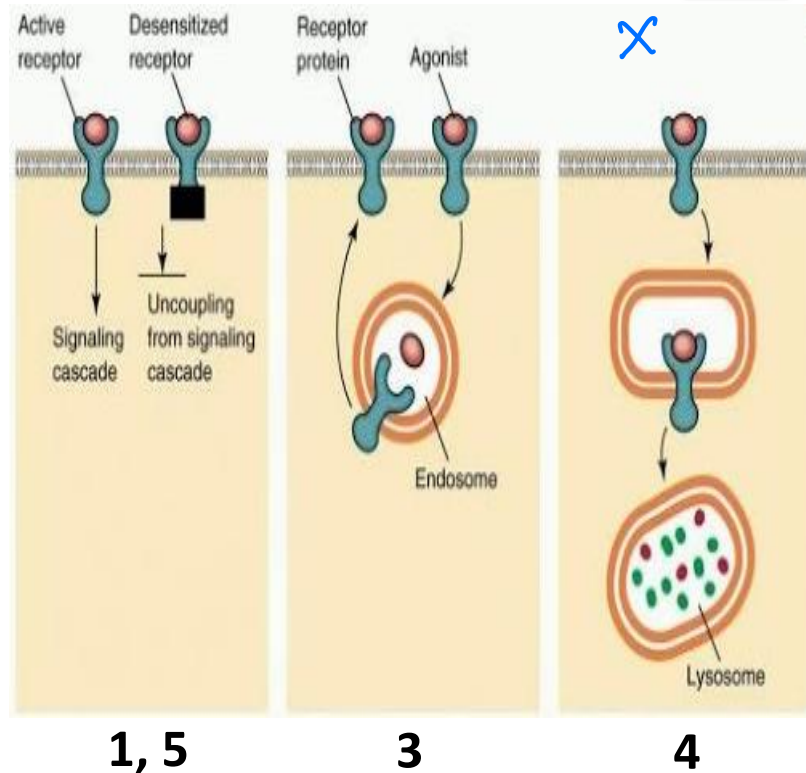
➤ Characters of cell receptors:

- The number of receptors in a target cell does not remain constant.
- **Increased** hormone **concentration** and increased **binding** with its target cell receptors cause the number of receptors to **decrease**.
- This is known as **down regulation** that decreases the target tissue's **responsiveness** to the hormone.



Hormones receptors structure, distribution & ligands

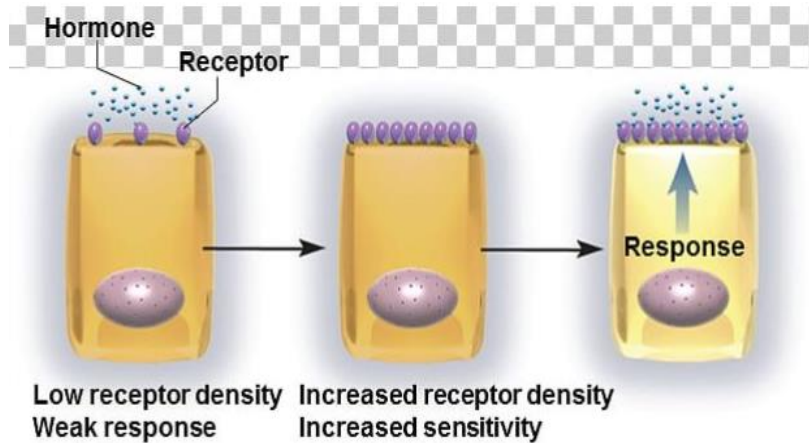
- Down-regulation can occur through:
 1. Inactivation of some receptor molecules.
 2. Decreased production of the receptors.
 3. Temporary sequestration of the receptor to the inside of the cell, away from the site of action of hormones.
 4. Destruction of the receptors by lysosomes after they are internalized.
 5. Inactivation of some intracellular protein signaling molecules.



Hormones receptors structure, distribution & ligands

➤ *Up-regulation* of receptors:

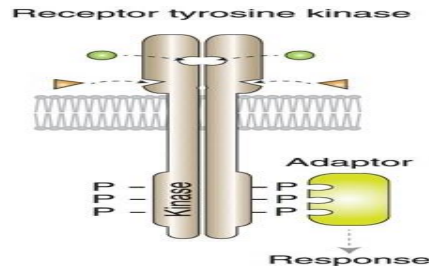
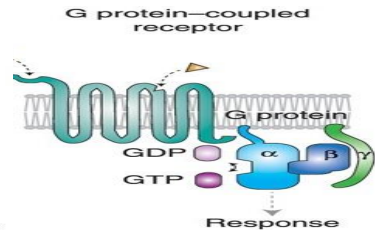
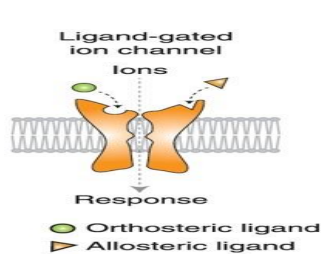
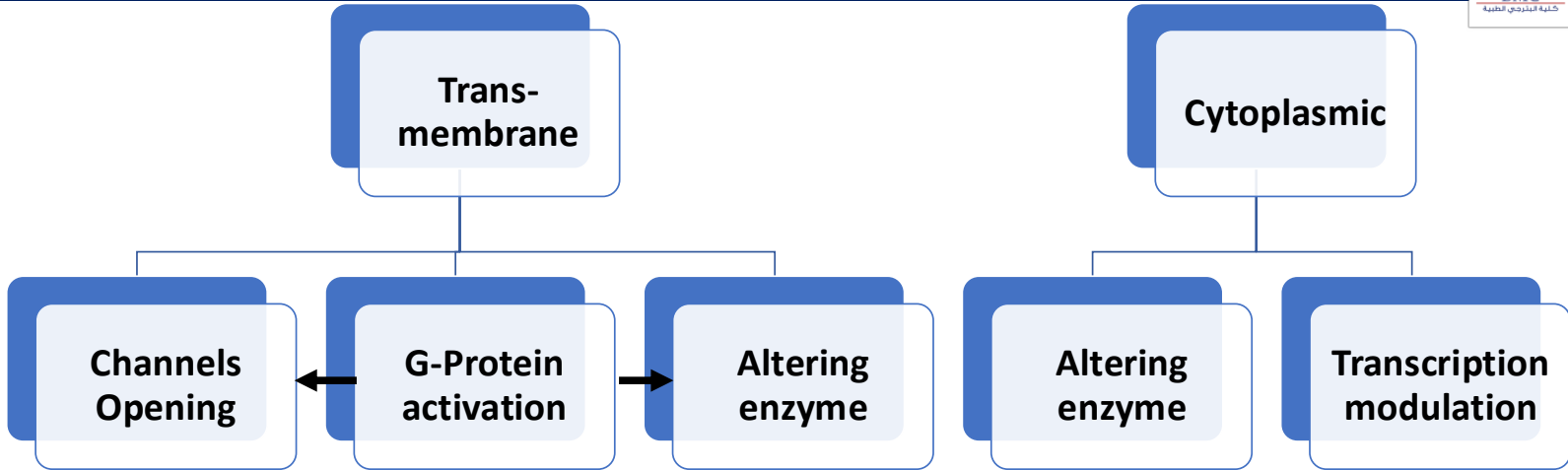
- It is **increased** cell response 2ry to prolonged exposure to **decreased** amounts of hormone.
- The target tissue becomes progressively **more sensitive** to the stimulating effects of the hormone
- This can happen either by **increase** in receptor **expression** or **decrease** in rate of **internalization** of receptors.





Hormones receptors intracellular signaling pathways

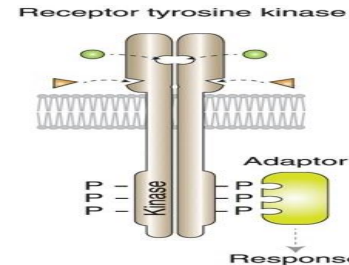
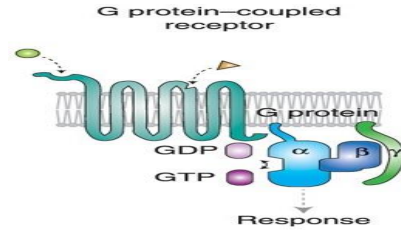
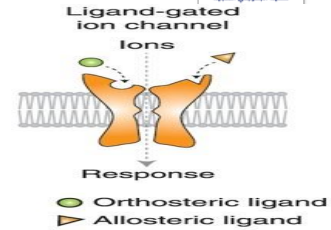
Hormones receptors intracellular signaling pathways



Hormones receptors intracellular signaling pathways

➤ The mechanisms by which chemical messengers exert their IC effects may be through:

- A. Opening of certain ion channels (trans-membrane).
- B. G-Protein activation (trans-membrane).
- C. Altering enzyme activity within the cell (trans-membrane or cytoplasmic).
- D. Activation of transcription (cytoplasmic or nuclear).



The resulting cell-signaling pathways provide:

- 1. **Amplification** (same) and **distribution** (different) of the effect.
- 2. **Regulation** and **feedback** control.

Thank You!



Conclusion

- **Hormones have different characteristics.**
- **Hormones may be protein, polypeptides, steroids or Tyrosine derivatives.**
- **Hormone receptors may be trans-membrane, cytoplasmic or nuclear.**

Case scenario

A researcher is studying cortisol, a steroid hormone secreted by the adrenal cortex. She wants to identify where its primary receptor is located within the target cell. Based on the known receptor location for steroid hormones, where would she expect to find it?

1. In the mitochondria
2. On the surface of the cell membrane
3. In the cell cytoplasm
4. In the cell nucleus, bound to chromosomes

References

Guyton and Hall, 13th edition. Unit II(6).

Ganong's review of medical physiology 25th ed. Section I(5).

Thank You